Classroom Assignment: Understanding JavaScript Scoping (var vs let vs global)

Learning Objective:

Understand **global scope, function scope, and block scope** by using var and let inside conditional blocks.

Expected Completion Time:

Best Case: *__15 minutes__*

Average Case: *__20 minutes__*

Assignment Details:

Write a JavaScript program to observe how the same variable name behaves:

* in *__global scope__*,

* inside a *__function__*, and

* inside an *__if-block__* using both var and let.

__Assignment Requirements__:

1. Declare a *__global variable__* named genderType with value "female".

2. Create a function named *__printGender__*.

3. Inside the function, declare a *__function-scoped__* variable color with value "brown" using let.

4. Create an *__if condition__* that checks whether genderType starts with "female".

5. Inside this if-block:

   * Declare a variable age = 30 using *__var__*.

   * Create a *__block-scoped__* variable color = "pink" using let.

   * Print the color inside the block and observe which value appears.

6. Outside the if-block but inside the function, print the value of age.

7. Call the function and print the value of genderType globally.

8. Now change the global variable named **genderType** with value “male” and observe the functionality of scoping in JavaScript.

Hints to Solve:

* var inside a block acts as if declared at the *function level*.

* let inside a block is visible *only inside that block*.

* Try printing color from different places to observe which value is picked.

* Compare the behavior of the same variable name "color" when declared with let inside function vs inside block.

Expected Outcome:

After completing this assignment, you should be able to

* Understand how *global variables* can be accessed inside functions.

* Differentiate *function-scope* (var) from *block-scope* (let).

* Explain why the color variable prints "pink" inside the block but remains "brown" outside.

* Understand why age is accessible outside the if-block.